Q-Relay: retrained A0–A3 ablations

Exploratory, already-observed twenty-scenario panel

Three independent training checkpoints per arm; each fixed-final run used 3,000 updates, 9M transitions, 11 physical workers and 33 logical environments. No early stopping, seed filtering, resume, or evaluation-based checkpoint choice. Shared scenario IDs 657201–657220 are paired across arms; sixty trajectories per arm are not sixty training replicates.

Interventions and retained components

A1 zeros only lag-difference history. A2 removes only public admission. A3 uses a parameter-free zero security critic and fixes the security multiplier to zero. Actor parameterization, cell diversity and retained critics/duals are unchanged. A0–A2 retain a distinct trainable security critic. A4 (no cell diversity) and A5 (shared critic) were not executed.

All twelve checkpoint outcomes

Arm	Training seed	Late TPS	Zero (%)	Failure (%)	Max / f=6
A0	657111	312.725	11.650	11.650	2
A0	657112	312.632	12.465	12.465	2
A0	657113	284.438	16.565	16.565	3
A1	657121	296.499	14.185	14.185	2
A1	657122	323.030	11.175	11.175	1
A1	657123	309.411	10.075	10.075	0
A2	657131	300.378	18.000	18.000	3
A2	657132	309.148	15.430	15.430	3
A2	657133	307.840	15.755	15.755	2
A3	657141	282.763	16.465	16.465	2
A3	657142	324.269	11.775	11.775	2
A3	657143	307.387	14.115	14.115	3

Late TPS averages steps 1,500–1,999 of 2,000-step deterministic prefix-conditioned argmax trajectories. Thirty failures occur at step 1,000 with cascading stress. Raw post-event zero TPS and failure are calculated separately; their percentages happen to coincide here. Actual global maximum exposure is retained; all twelve checkpoints have zero violating steps and no forced-zero TPS.

Equal-weight arm summaries and interpretation

Arm	Mean late TPS	Seed SD (n=3)	Zero (%)	Actual max
A0	303.265	16.305	13.560	3
A1	309.647	13.267	11.812	2
A2	305.789	4.731	16.395	3
A3	304.806	20.873	14.118	3

Primary late-TPS contrasts against A0

Contrast	Difference	Hierarchical 95% interval	Holm p
A1 – A0	+6.382	[-15.114, 28.226]	0.149031
A2 – A0	+2.524	[-13.260, 22.175]	0.884659
A3 – A0	+1.541	[-24.462, 28.650]	0.884659

Analysis was registered before the final extension run, after this panel had already been observed. Ten thousand hierarchical bootstrap replicates independently resample training checkpoints within arms and pair scenario IDs across arms (RNG 99111). Exact two-sided sign flips enumerate all 2^20 assignments of training-averaged paired scenario differences. Holm covers only the three primary contrasts above.

Negative and limited findings

Every hierarchical interval includes zero; no primary contrast is Holm-significant. Removing history did not produce the predicted decrease: A1 is descriptively +6.382 TPS above A0. Removing admission increases mean zero service from 13.560% to 16.395%, a descriptive secondary result. A3 remains at maximum exposure 3 with zero violations, which does not establish that a security critic is unnecessary in other settings. All three full-checkpoint late means exceed the best-TPS baseline CMAB (252.912), descriptively and on this panel only.

Frozen stochastic streams

For the original A0 checkpoint, three already-recorded stochastic streams per scenario give stochastic-minus-deterministic late TPS −25.308, conditional paired 95% interval [−39.137, −11.929], and zero-service +2.333 percentage points [0.428, 4.485]. This new read-only 10,000-resample summary uses RNG 99113; action streams are not training seeds.

Reproduction evidence

The accompanying ablation_results.csv contains 240 rows; checkpoint_results.csv and checkpoint_manifest.csv retain all twelve checkpoints, budgets, source/config hashes and receipts. Selected-switch JSON/patch files are retrospective readable summaries, not replacement prospective registrations. Original frozen plans, negative baselines, failures and sealed audits are preserved. No population training-seed significance or unseen-panel claim is made.